

- 8 (a) In plants such as rice, *Oryza sativa*, rubisco has a low rate of activity, which in turn affects the rate of photosynthesis. The cereal crop sorghum, *Sorghum bicolor*, has a high rate of activity of rubisco.

A genetically modified (GM) variety of rice was produced. Parts of the quaternary structure of rubisco in rice were altered to be the same as the rubisco in sorghum.

The rate of activity of rubisco in non-GM rice and GM rice was measured at different concentrations of atmospheric carbon dioxide (CO_2).

The results are shown in Fig. 8.1.

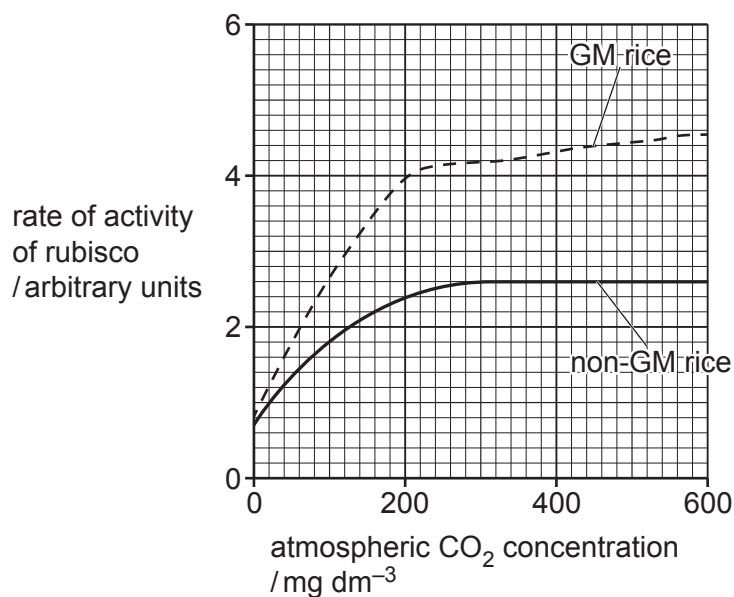


Fig. 8.1

- (i) Compare the two curves shown in Fig. 8.1 and explain why the curve for non-GM rice levels off.

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..... [3]

- (ii) Suggest which part of the rubisco molecule was altered to produce the GM variety of rice.

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..... [1]

(b) Fig. 8.2 shows an outline of the Calvin cycle.

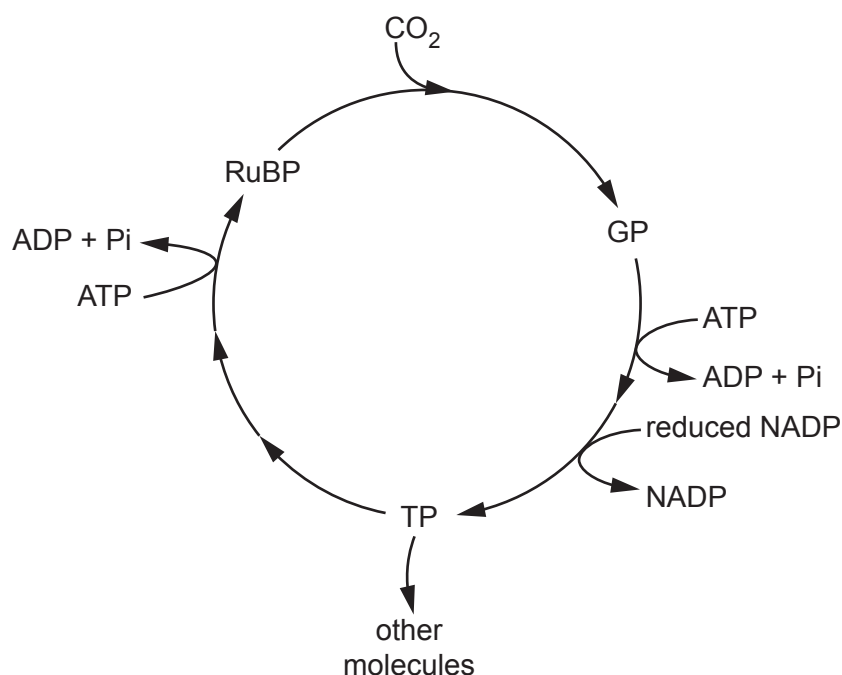


Fig. 8.2

(i) Name the process that involves the enzyme rubisco.

..... [1]

(ii) State the type of reaction that occurs when GP is converted to TP .

..... [1]

(iii) RuBP is regenerated from molecules of TP .

State how many molecules of RuBP are produced from 10 molecules of TP .

..... [1]

(iv) TP molecules that are **not** involved in the regeneration of RuBP can be used in the synthesis of other molecules.

State **two** molecules that can be produced from these TP molecules.

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 [2]

[Total: 9]